Project 2: Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

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Outline

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2. **State of the Art**
3. **Proposed Method**: Extension of DRCTConvB with frequency-domain information and multi-head joint learning.
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Problem Statement

Why the necessity of detecting post-processing AI-images?

AI-Generated Image detectors are usually trained and evaluated in idealized conditions.

In real-world scenarios, images are commonly compressed and transformed by social media platforms.

These transformations can alter image patterns and introduce artifacts, leading the detectors in error.

This is especially critical for AI-generated fake images intended to influence public opinion.

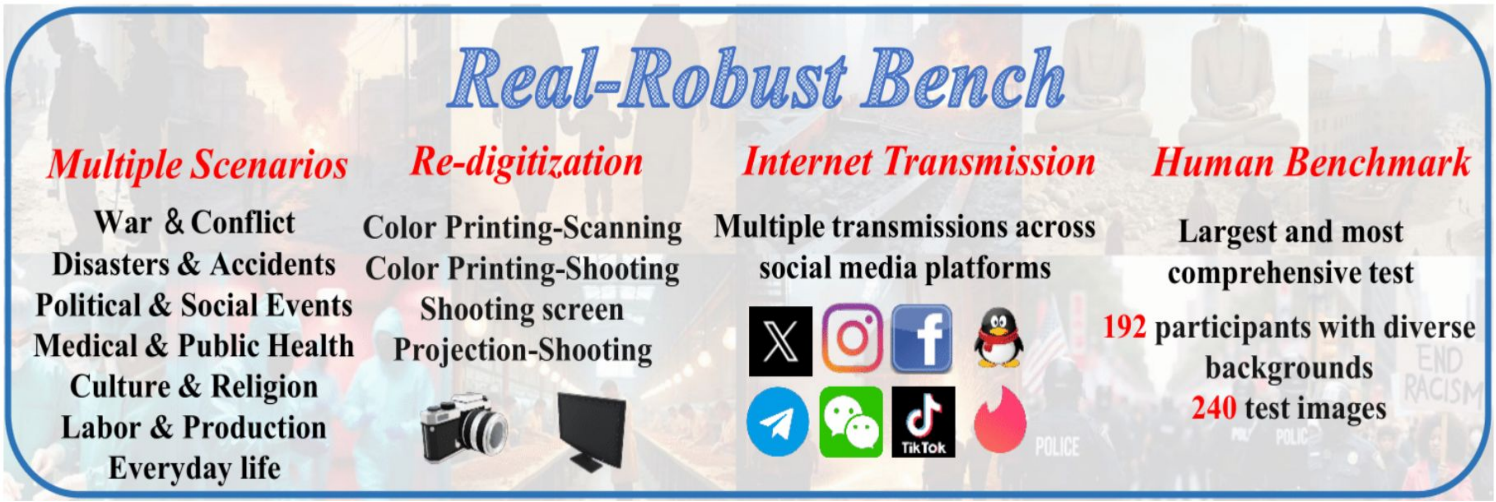
State of the Art

RRDataset/RRBench

A benchmark containing images with different transformation categories

Several state-of-the-art AI-detectors were fine-tuned on this dataset by authors

The benchmark shows that post-processing transformations can significantly reduce real/fake detection performances.



DRCTConvBase (Our Baseline)

DRCT improves AI-generated image detection through diffusion reconstruction and contrastive training.

DRCTConvB was reported as the strongest detector among the considered models when fine-tuned and used in real world scenario.

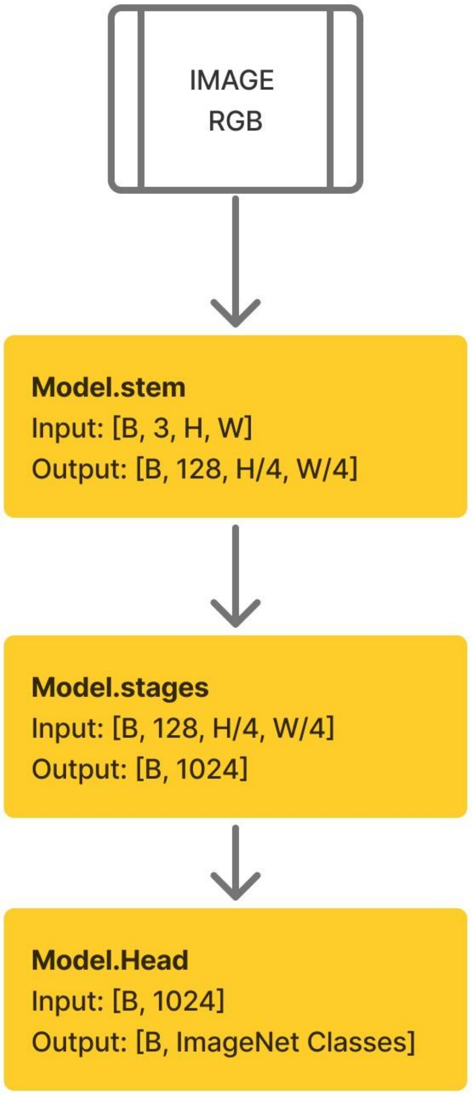
However, its performance still drops under re-digitalization transformations.

Model	Original		Transmission		Re-digitization		Overall
	Fake(%)	Real(%)	Fake(%)	Real(%)	Fake(%)	Real(%)	ACC(%)
Detectors (Train on GenImage-SDv1.4 & fine-tune on RRDataset)							
DRCT-ConvB[8]	93.52	95.52	92.82(-0.70)	95.09(-0.43)	64.34(-29.18)	96.22(+0.70)	89.59

ConvNeXt

DRCTConvB is based on the ConvNeXt family.

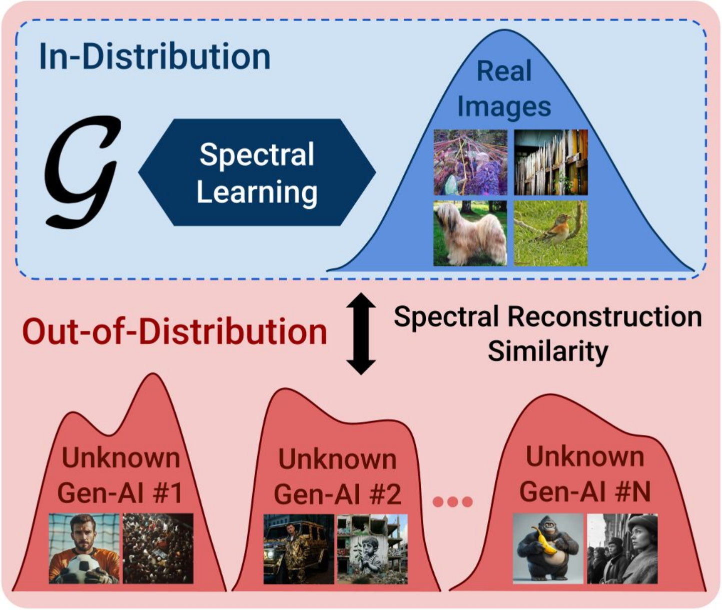
They modernize ResNet CNNs by including and adapting design choices and ideas inspired by Vision Transformers, while preserving the simplicity of convolutional networks.



SPAI

SPAI models the spectral distribution of real images and detects AI-generated images when the spectrum is out-of-distribution.

This supports the idea that frequency-domain representations can capture artifacts not easily visible in the RGB domain.

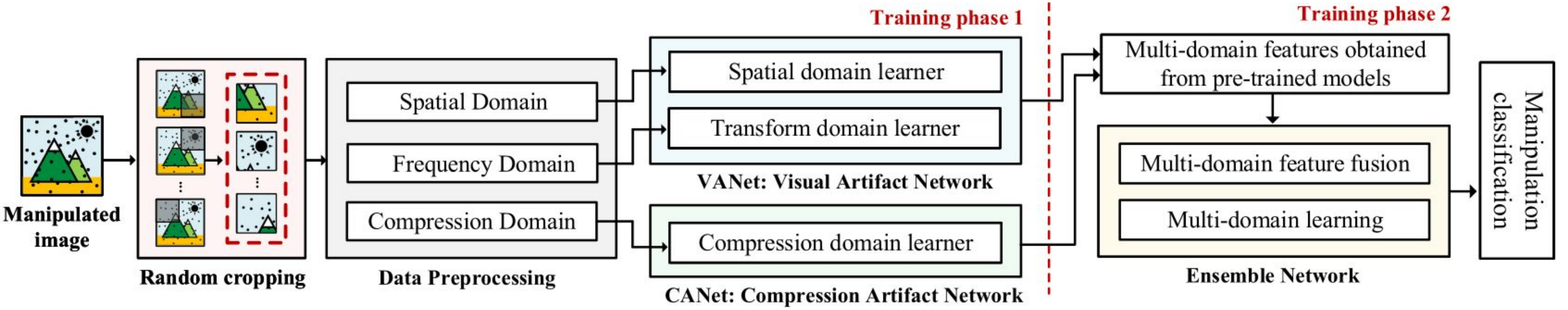


MCNet

Image forensics has also studied the classification of post-processing operations.

MCNet classifies JPEG image manipulations by combining spatial, frequency, and compression-domain features.

However, this line of work focuses on manipulation classification, not on jointly predicting AI/real class and transformation category.



Proposed Method

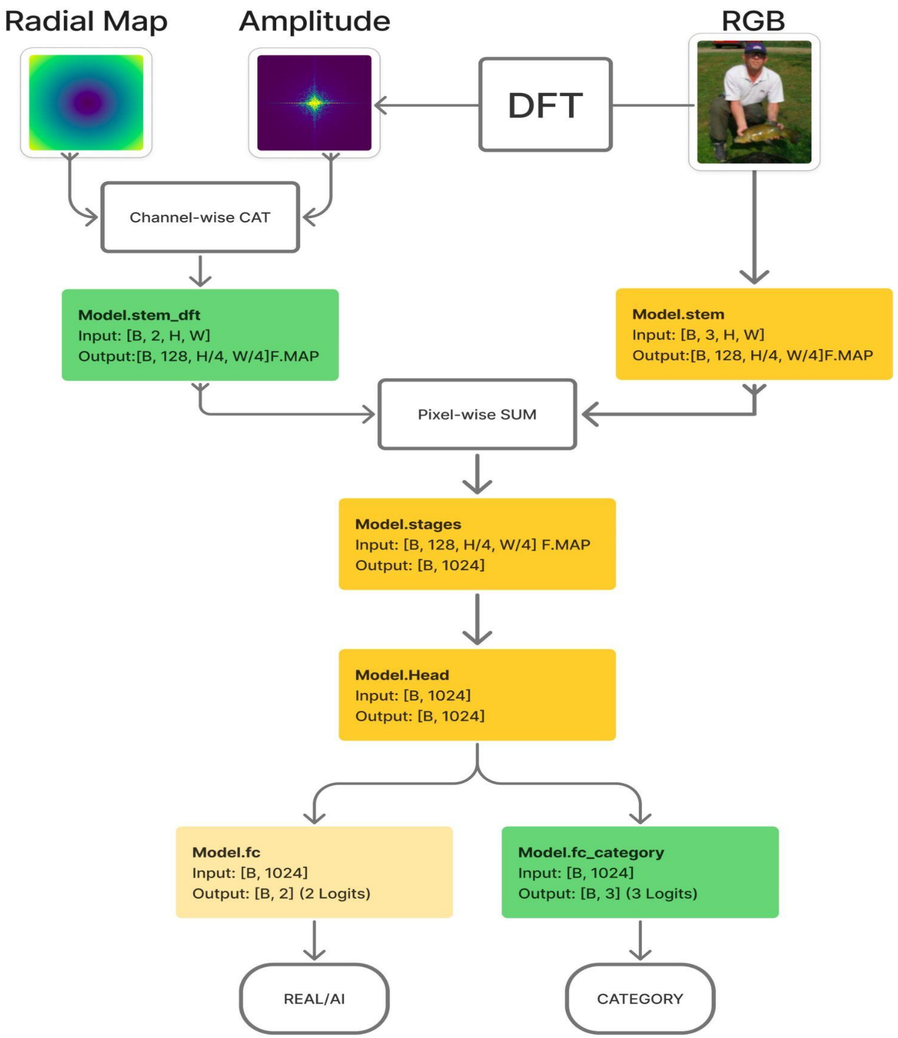
Base Idea

Instead of relying only on RGB spatial features, we add an initial parallel DFT Stem branch to DRCTConvB in order to use frequency-domain informations. We named it DRCTConvB-DFT

Some post-processing transformations and AI-generations can leave detectable traces in the frequency spectrum (acting like a low-pass or high-pass filter).

Explore a multi-task extension of the model that jointly predicts image authenticity and transformation category.

DRCTConvBase-DFT



- The added blocks are marked in green
- A radial map, that is an image containing in each pixel the normalized distance from the center, is concatenated channel-wise with the DFT amplitude, in order to help the CNN distinguish low and high-frequency regions.

Terminology

Task name	Description	Number of classes
Class task	Real vs AI-generated image classification	2
Category task	Post-processing transformation classification	3

Combined Loss Function

Normalize each task loss by its own uniform probability prediction error with K classes:

$$CE_{\text{norm}} = \frac{CE(y, \hat{y})}{\log(K)}$$

With this normalization, a model with uniform prediction gives:

$$CE_{\text{norm}} = 1.$$

The combined weighted loss used for the joint tasks training is given by:

$$L = \frac{w_{\text{AI / R}} \cdot CE_{\text{class, norm}} + w_{\text{cat}} \cdot CE_{\text{cat, norm}}}{w_{\text{AI / R}} + w_{\text{cat}}}$$

Dataset

Subset Creation

We trained the models in the Google Colab Environment (T4 and A100 GPUs)

Then it was impractical to use the whole RRDataset

We took 1,500 images for each big group:

- original/ai
- original/real
- redigital/ai
- redigital/real
- transfer/ai
- transfer/real

Then each macro-category is divided into three splits. This grants balance

Experimental Setup

Experimental Questions

1. Does the parallel DFT Stem branch improve real/AI detection?
2. Does the DFT branch help in recognizing transformations?
3. Does category classification benefit from representations and embeddings learned for real/Ai detection task?
4. Does joint multi-head training improve both tasks?
5. How do different combinations of loss weights affect the trade-off between the two tasks?

Starting Checkpoints

Three starting checkpoints used during the experiments:

Checkpoint	Description
DRCT Base Checkpoint	Original DRCTConvB checkpoint before fine-tuning on our Dataset.
RGB-Only Class Task Checkpoint	DRCTConvB fine-tuned on our Dataset in the class task only.
DFT Class Task Checkpoint	DRCTConvB-DFT fine-tuned on our Dataset in the class task only.

Loss-Weight Ablation

$w_{\mathbf{AI/R}}$	$w_{\mathbf{cat}}$	**AI/R contribution**	**Category contribution**
1	0.25	80.00%	20.00%
1	1	50.00%	50.00%
1	1.75	36.36%	63.64%
1	2.7665	26.55%	73.45%
1	5	16.67%	83.33%

Table 1: Loss weights used in the experiments and their nominal relative contribution to the total combined loss.

Model Evaluation

Experiment 1

Experiment 1 (1)

Single-Head class task training

We compared:

- DRCTConvB model fine-tuned on our Dataset;
- DRCTConvB-DFT model fine-tuned on our Dataset.

Network Type	Accuracy on Val set	Accuracy on Test set	Loss on Val set	Loss on Test set
DRCTConvB	0.7874	0.8311	**0.5417**	**0.4293**
DRCTConvB with Stem DFT	**0.8163**	**0.8504**	0.5498	0.4901

Table 2: Comparative performance of our proposed model (highlighted in blue) and the DRCT detector on Single Head Real/AI classification task

Experiment 1 (2)

1. Does the parallel DFT Stem branch improve real/AI detection?

YES

The outputs of this experiment are the two anticipated checkpoints:

Checkpoint	Description
DRCT Base Checkpoint	Original DRCTConvB checkpoint before fine-tuning on our Dataset.
RGB-Only Class Task Checkpoint	DRCTConvB fine-tuned on our Dataset in the class task only.
DFT Class Task Checkpoint	DRCTConvB-DFT fine-tuned on our Dataset in the class task only.

Experiment 1 (3)

By looking at the AI/Real accuracy for each transformation category, the models perform better in redigital category while perform worse on the transfer category (in particular our proposed model)

Category	AI	Real	Overall
Original	85.33	77.78	81.56
Redigital	87.56 (+2.23)	82.67 (+4.89)	85.11 (+3.55)
Transfer	**83.56** (-1.77)	**81.78** (+4.00)	**82.67** (+1.11)

Category	AI	Real	Overall
Original	**88.00**	**83.56**	**85.78**
Redigital	**88.89** (+0.89)	**85.78** (+2.22)	**87.33** (+1.55)
Transfer	**83.56** (-4.44)	80.44 (-3.12)	82.00 (-3.78)

Experiment 2

Experiment 2 (1)

Semi-Multi-Head category task training starting from checkpoints obtained after the previous experiment.

We compared:

- DRCTConvB model
- DRCTConvB-DFT model

Multihead Network Type	Class Acc	Category Acc	Class Loss	Category Loss	Class F1-macro	Category F1-macro
DRCTConvB	0.7837	0.5593	**0.7118**	**0.7406**	0.7807	0.4680
DRCTConvB with DFT Stem	**0.8326**	**0.7322**	1.0973	1.1122	**0.8325**	**0.7332**

Experiment 2 (2)

2. Does the DFT branch help in recognizing transformations?

Yes

From these first two experiments we can confirm that our proposed model is better than the base DRCT detector in both tasks, and in particular frequency-domain informations are especially useful for recognizing transformations.

From now on we will consider only DRCTConvB-DFT

Experiment 3

Experiment 3 (1)

Single Head category task training

Studied if the category task benefits from representations previously learned for real/AI image detection across transformations.
We compared two different starting points for DRCTConvB-DFT:

- from the DRCT Base Checkpoint;
- from the DFT Class Task Checkpoint;

DFT Starting Checkpoint	Class Acc	Category Acc	Class Loss	Category Loss	Class F1-macro	Category F1-macro
DFT CLASS TASK CHECKP.	0.5063	**0.8115**	2.7369	**0.4617**	0.4946	**0.8141**
DRCT BASE CHECKPOINT	-	0.5626	-	0.7880	-	0.5264

Experiment 3 (2)

3. Does category classification benefit from representations and embeddings learned for real/Ai detection task?

Yes

This motivates the use of a joint multi-head training combined loss.

Experiment 4

Experiment 4 (1)

Comparison between joint multi-head training and the corresponding single-head trainings, starting from the same checkpoint (DRCT Base Checkpoint)

Task Type	Class Acc	Category Acc	Class Loss	Category Loss	Class F1-macro	Category F1-macro
Combined MultiH	**0.8767**	**0.8409**	0.5668	**0.4111**	**0.8765**	**0.8404**
Category SingleH	-	0.5626	-	0.7880	-	0.5264
Class SingleH	0.8334	-	**0.5200**	-	0.8332	-

Experiment 4 (2)

4. Does joint multi-head training improve both tasks?

Yes

Joint training improved both class and category tasks, suggesting that both tasks can benefit from each other and provide complementary supervision when optimized together.

Experiment 4 (3)

The multi-head model is stronger on real image detection across transformations, while the single-head model performs better on AI images.

Category	Model	AI	Real	Overall
Original	Class SingleH	**88.00**	83.56	85.78
	Combined MultiH	80.00	**93.78**	**86.89**
Redigital	Class SingleH	**88.89** (+0.89)	85.78 (+2.22)	87.33 (+1.55)
	Combined MultiH	80.89 (+0.89)	**94.22** (+0.44)	**87.56** (+0.67)
Transfer	Class SingleH	83.56 (-4.44)	80.44 (-3.12)	82.00 (-3.78)
	Combined MultiH	**86.22** (+6.22)	**91.56** (-2.22)	**88.89** (+2.00)

Experiment 5

Experiment 5 (1)

Studied how the relative loss weights affect the trade-off between Real/AI detection and category classification

We kept w(AI/R)=1 fixed and varied w(cat).

Loss Cat Weight	Class Acc	Category Acc	Class F1-macro	Category F1-macro
$w_{\mathrm{cat}} = 0.25$	0.9256	0.8904	0.9256	0.8912
$w_{\mathrm{cat}} = 1.00$	**0.9293**	0.8915	**0.9292**	0.8921
$w_{\mathrm{cat}} = 1.75$	0.9089	0.9063	0.9086	0.9064
$w_{\mathrm{cat}} = 2.7665$	0.9237	0.8982	0.9237	0.8981
$w_{\mathrm{cat}} = 5.00$	0.9097	**0.9167**	0.9096	**0.9167**

Table 17: Comparison of multi-head fine-tuning performance averaged between the VALIDATION SET and TEST SET, using different values of w_{cat} while keeping $w_{\mathrm{AI/R}} = 1$ fixed

Experiment 5 (2)

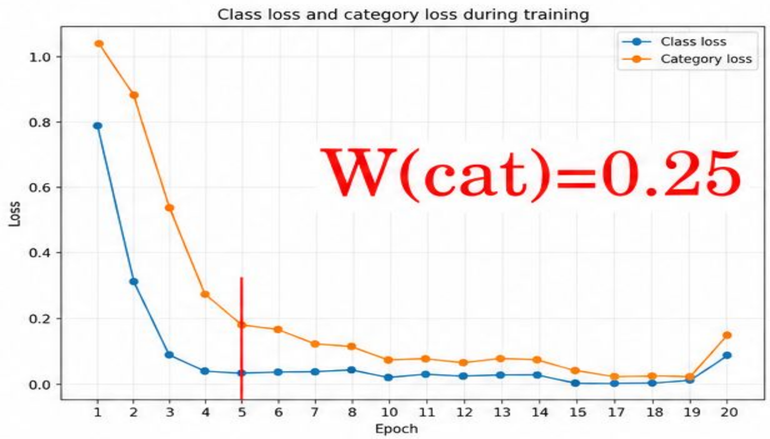
5. How do different combinations of loss weights affect the trade-off between the two tasks?

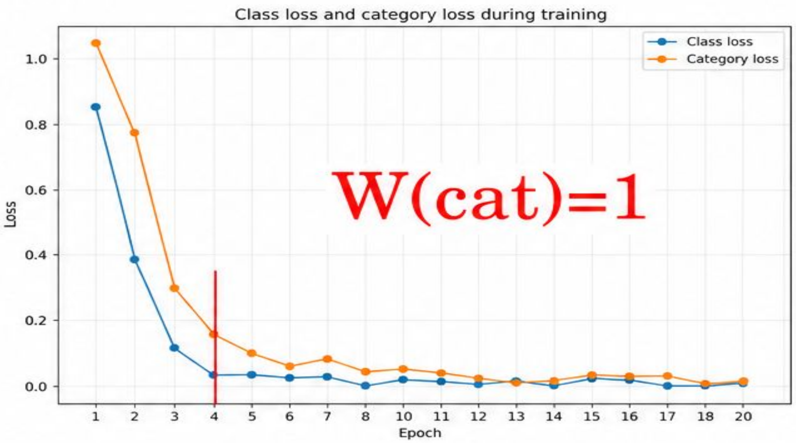
The differences between configurations are not extremely large, but we can extract these practical guidelines:

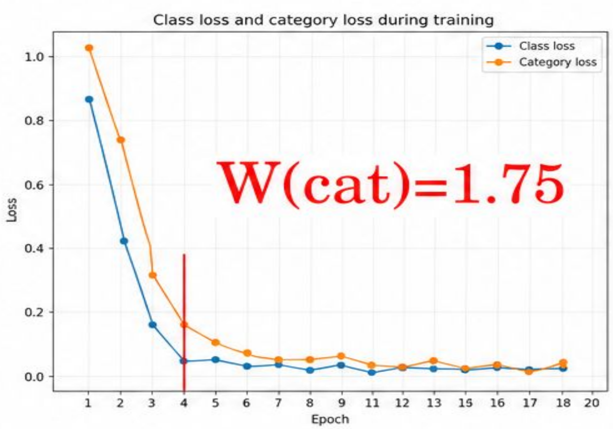
Main objective	$w_{\mathrm{AI/R}}$	w_{cat}
AI/Real detection	1	1
Balanced trade-off between AI/Real and category classification	1	2.7665
Category classification	1	5

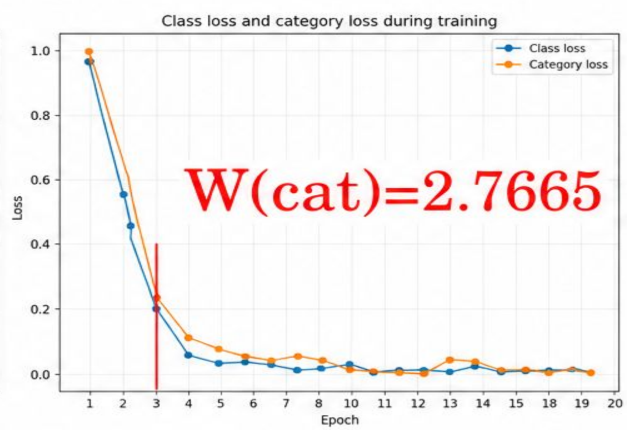
Table 1: Recommended loss-weight configurations according to the main optimization objective.

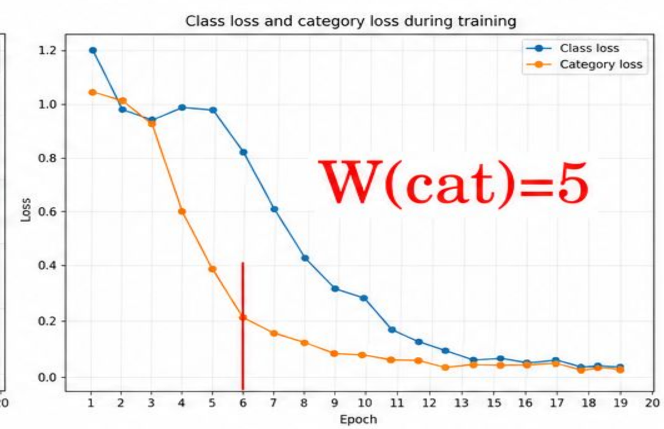
Experiment 5 (3)











The marked epoch shows when the category loss reaches 0.20.

Experiment 5 (4)

Regardless of the nominal weight assigned to the category loss term, it follows the learning dynamics of the real/AI detection task.

This confirms again that the category task strongly depends on the shared representation learned with the class task, and it is not driven only by its explicit loss.

Experiment 5 (5)

Across weighting configurations there is a recurrent pattern: models detect AI-gen images better after redigital and transfer transf., but this comes with a drop in real-image detection for those categories.

Category	w_{cat}	AI	Real	Overall
Original	0.25	88.89	94.67	91.78
	1	91.11	95.56	**93.33**
	1.75	84.00	**98.22**	91.11
	2.7665	**93.33**	92.89	93.11
	5	92.89	92.89	92.89
Redigital	0.25	95.11 (+6.22)	93.33 (-1.34)	**94.22** (+2.44)
	1	92.00 (+0.89)	92.44 (-3.12)	92.22 (-1.11)
	1.75	92.44 (+8.44)	**94.22** (-4.00)	93.33 (+2.22)
	2.7665	**97.78** (+4.45)	87.11 (-5.78)	92.44 (-0.67)
	5	89.33 (-3.56)	**94.22** (+1.33)	91.78 (-1.11)
Transfer	0.25	**96.44** (+7.55)	95.11 (+0.44)	**95.78** (+4.00)
	1	95.11 (+4.00)	93.78 (-1.78)	94.44 (+1.11)
	1.75	87.11 (+3.11)	**96.89** (-1.33)	92.00 (+0.89)
	2.7665	95.11 (+1.78)	88.89 (-4.00)	92.00 (-1.11)
	5	94.22 (+1.33)	88.44 (-4.45)	91.33 (-1.56)

Conclusion

- Our proposed model, DRCTConvB with a parallel DFT Stem, has generally better performance than the base DRCTConvB in the detection of AI/Real Images and in the classification of post-processing transformations.
- Jointly training the two tasks can achieve better results in detection of AI images and post-processing classification compared with the corresponding single-task models.

Thank you for your attention!

References

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